Micromechanical Investigation of soil deformation: Incremental response and granular ratcheting

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(July 28, 2006)

The mechanical behavior of soils has been largely investigated using constitutive models. These are empirical relations based on laboratory tests of soil specimens. The investigation of the soils at the grain scale using discrete element models has become possible in recent years. These models have provided valuable understanding of many micromechanical aspects of soil deformation The aim of this work is to draw together these two approaches in the investigation of the plastic deformation of soils. A simple discrete element model has been used to evaluate the effect of anisotropy, force chains, and sliding contacts on different aspects of soil plasticity: dilatancy, failure, shear bands, ratcheting, etc. The discussion of these aspects raises important questions such as the width of shear bands, the origin of the stress-dilatancy relation, and the existence of a purely elastic regime in the deformation of granular materials.

1 Introduction

The 1960s was significant for the development of soil mechanics and, in particular, the constitutive models for soils. Prior to this decade, soil mechanics was confined to linear elastic theory and the Mohr-Coulomb failure criterion. A radical change of the perspectives of soil plasticity occurred after the pioneering work of Roscoe and his coworkers in Cambridge, which led to the basic principles of the Critical State Theory [1,2]. In an attempt to cover further aspects of cyclic soil behavior, subsequent developments have given rise to a great number of constitutive models [3]. Unfortunately several models only give satisfactory results in the small range of experiments where they were developed. Other models attempting to represent a wider range of phenomena had to incorporate a large number of parameters. In most cases these parameters lack physical meaning, and they are difficult to calibrate with the experimental data [4].

In geotechnical applications it is desirable that the parameters of a constitutive relation depend directly on the properties of the grains. In the simple case of dry soils, granulometric properties can involve grain shape and angularity, distribution of grain sizes, friction coefficient and stiffness of the grains [5]. The development of micromechanical constitutive models has been specially motivated by recent investigations on granular materials at grain scale [6]. Numerical and laboratory experiments show that stress in granular materials transmitted through an heterogeneous contact network with broad force distribution [7] . Under small deviatoric loads, an initially isotropic packing develops an anisotropic contact network because new contacts are created along the loading direction, while some contacts are lost perpendicular to it. This geometrical anisotropy leads in turn to an anisotropic response of the granular assembly, whose effect on the anisotropic elasticity and the plasticity remains an open issue [8–10].

In this work we perform a micromechanical investigation of the elastoplastic response of granular materials. Granular samples are represented by assemblies of polygons generated by Voronoi tessellation. This method gives a diversity of areas of polygons following a Gaussian distribution with mean value ℓ^2 and variance of $0.36\ell^2$. The number of edges of the polygons is distributed between 4 and 8 for 98.7% of the polygons, with a mean value of 6. The interparticle forces include elasticity, viscous damping and friction with the possibility of slip. The ratio between the tangential and normal contact stiffnesses is $k_t/k_n = 0.33$, and the friction coefficient is $\mu = 0.25$. In Sections 2 and 3 we investigate shear band formation and incremental response using perfect polygonal packings. The polygonal particles have initially exactly adjusted

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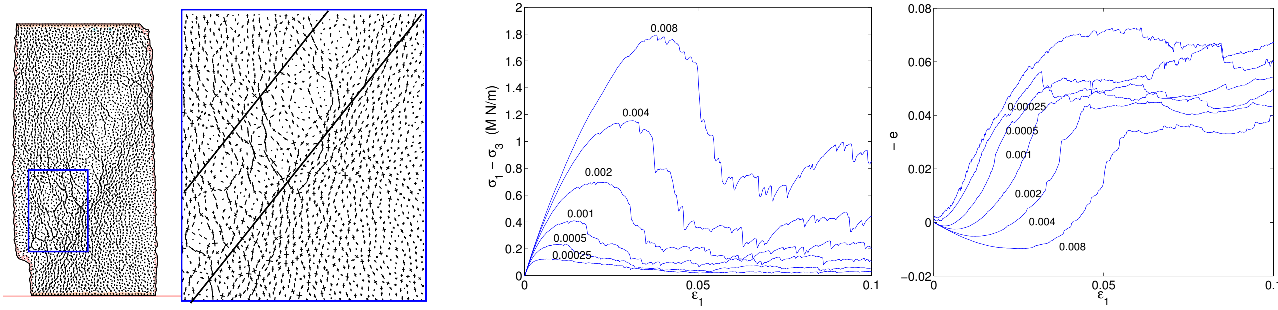

Figure 1. From Left to right: Principal stress directions of the grains after failure ($\epsilon_1 = 0.07$); the confining pressure is $p_0 = 0.001k_n$. Detail of the stress in the shear band. Deviatoric stress and volumetric strain versus axial strain for different values of p/k_n, where p is the lateral pressure. $e > 0$ represent compression of the sample.

shapes and leave null porosity such as the case of dry masonry walls or marble [11]. In Section 4 we introduce the granular ratcheting, as the response of a dense polygonal packing when they are subjected to load-unload cycles.

2 Shear band formation

The presence of shear bands as a failure mode is very sensitive to the boundary conditions. We chose them in order to mimic the experimental tests under plane strain conditions: First, a confining pressure is applied to the sample through a flexible membrane. Then, two horizontal walls at the top and bottom of the packing are used to apply vertical loading with constant velocity [12].

The deformation of the assembly involves creation and loss of contacts as well as restructuring by means of rolling and sliding contacts. These changes imply a continuous variation of the stress-strain relation and a change of the void ratio during load. The stress tensor is calculated from the forces applied on the boundary of the sample as $\sigma_{ij} = \frac{1}{A}\sum_b f_i^b x_j^b$, where $\vec{x}^b$ is the point of application of the boundary force $\vec{f}^b$ and A is the area enclosed by the membrane. From the principal values of this tensor, one can define the mean normal stress $p = (\sigma_1 + \sigma_2)/2$ and the deviatoric stress $q = (\sigma_1 - \sigma_2)/2$. The axial strain is calculated as $\epsilon_1 = \Delta H/H_0$, where H is the height of the sample. The volumetric strain is given by $\epsilon_V = \Delta A/A_0$.

The dependence of the deviatoric stress and the volumetric strain on the axial strain are shown in Fig. 1 for different confining pressures. A continuous decrease of the initial slope of the stress-strain curve is observed. Loading rearranges the contact network by means of sliding contacts, which in turn reduces the stiffness of the material. The initial compaction turns gradually to dilatancy. This transition is caused by loss of contacts perpendicular to the load direction, allowing the contact network to rearrange itself and inducing large plastic deformations. Near failure, the amount of plastic deformation is much larger than the elastic one. This considerably reduces the stiffness with respect to its initial value and makes the sample potentially unstable.

Plastic deformation by means of sliding contacts turns out to be a precursor mechanism of strain localizations. The frictional dissipation is uniformly distributed at the beginning of the load and it tends progressively to localize in thin layers, which ends up with the shear band formation [12]. Near failure, the orientational distribution of sliding contacts has its maximal value between the Mohr-Coulomb angle and the Roscoe angle, but rather closer to the former [13]. The shear band is given by a $6-8$ grain diameters thick layer where the frictional dissipation is more intense than on the average.

The characteristic width of the shear of the band can be associated to the propagation of stress inside the grains. The stress tensor at each particle P is given by $\sigma_{ij}^P = \frac{1}{a}\sum_c f_i^c \ell_j^c$ where a is the area of the polygon, f_i^c is the contact force and ℓ_j^c is the branch vector, connecting the center of mass of the polygon to the point of application of the contact force. The sum goes over all the contacts of the particle. The principal stress direction at each grain is represented in Fig. 1 by a cross. The length of the lines represents how large the components are. At the beginning of the loading, the major principal stress is almost parallel to the load direction, forming column-like structures which are called force chains. At failure these chains start buckling. The buckled chains gradually creates force loops which concentrate as shear bands. The size of such loops corresponds to the shear band width, and it depends only on the grain diameter. Buckling of each force chain involves rolling between the grains belonging to it, a feature which can be used to provide a theoretical explanation of the finite width of shear bands [14].

3 Incremental response

The method we use to calculate the strain response is the same as used in sand experiments [15]. It was introduced by Bardet [16] in the calculation of the incremental response using discrete element methods. We denote the incremental strain as $d\tilde{\epsilon} = (de, d\gamma)$, being e and γ the volumetric and shear components. The stress is represented by the vector $\tilde{\sigma} = (p, q)$, where p and q are the pressure and shear stress. The Bardet method will be used to determine the elastic and plastic components of the incremental strain as function of the stress state and the incremental stress $d\tilde{\sigma}$. First, the sample is isotropically compressed until it reaches the stress value $\sigma_1 = \sigma_3 = p - q$. Then, it is subjected to axial loading in order to increase the axial stress σ_1 to $p + q$. Loading the sample from $\tilde{\sigma}$ to $\tilde{\sigma} + d\tilde{\sigma}$ the strain increment $d\tilde{\epsilon}$ is obtained. Then the sample is unloaded to $\tilde{\sigma}$ and one finds a remaining strain $d\tilde{\epsilon}^p$, that corresponds to the plastic incremental strain. This procedure is implemented on different *clones* of the same sample, choosing different stress directions and the same stress amplitude in each one of them. We assume that the strain response after a reversal loading is completely elastic. Numerical simulations show that this assumption is not strictly true, because sliding contacts are always observed during the unload path [17, 18]. However, for stress amplitudes of $|d\tilde{\sigma}| = 0.001p$ the plastic deformation during the reversal stress path is less than 1% of the corresponding value of the elastic response. Within this margin of error, the difference $d\tilde{\epsilon}^e = d\tilde{\epsilon} - d\tilde{\epsilon}^p$ can be taken as the elastic component of the strain.

Fig. 2 shows the load-unload stress paths and the corresponding strain response when an initial stress state with $\sigma_1 = 1.25 \times 10^{-3} k_n$ and $\sigma_3 = 0.75 \times 10^{-3} k_n$ is chosen. The end of the load paths in the stress space maps into a strain envelope response $d\tilde{\epsilon}(\theta)$ in the strain space. Likewise, the end of the unload paths maps into a plastic envelope response $d\tilde{\epsilon}^p(\theta)$. This envelope consists of a very thin ellipse, nearly a straight line, which confirms the unidirectional aspect of the irreversible response predicted by the elastoplasticity

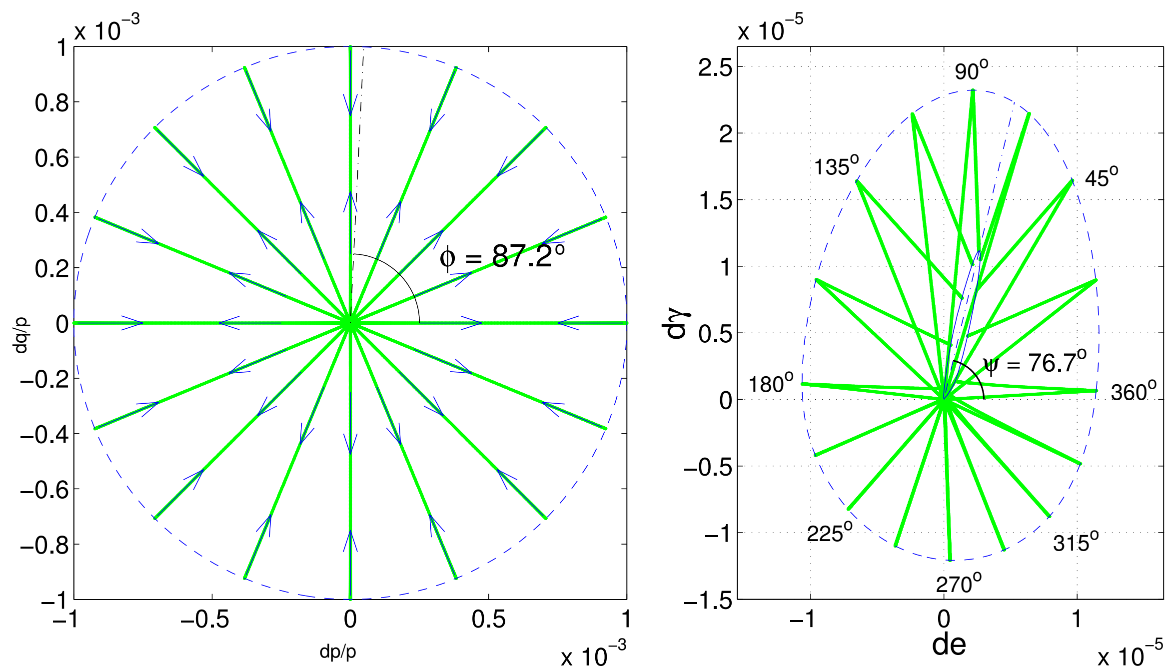

Figure 2. Stress - strain relation resulting from the load - unload test. Grey solid lines are the paths in the stress and strain spaces. Grey dash-dotted lines represent the yield direction (left) and the flow direction (right). Dashed line shows the strain envelope response and the solid line is the plastic envelope response.

theory [19]. The *yield direction* ϕ can be found from this response, as the direction in the stress space where the plastic response is maximal. In this example, this is around $\phi = 87.2°$. The *flow direction* ψ is given by the direction of the maximal plastic response in the strain space, which is around $76.7°$. The fact that these directions do not agree reflects a *non-associated flow rule*, that is also observed in experiments on realistic soils [15]. From numerical simulations of packings of disks, Bardet concluded also that a non-associated flow rule describes satisfactorily the incremental response [16]. This conclusion is also supported by several experimental tests on plane strain deformation [2, 19, 20]. Both numerical and experimental results show clearly deviations from the normality condition. This is connected to the fact that any load involves sliding contacts so that the elastic regime is vanishing small but not a finite domain as the elastoplasticity establishes [21]. Recent numerical simulations of three dimensional packings of spheres contradict not only the normality postulate [22], but also the unidirectionality of the flow rule [23], leading also to the conclusion that a profound modification of the elastoplasticity theory is required [24].

The elastic part of the incremental response is described using two Young moduli and two Poisson ratios [11]. The dependence of these variables on the stress is different from loose granular packings. We note that this is due to the fact that we start with a polygonal packing with zero porosity, where the force distribution is unusually narrow [13]. This is not typical in most granular materials where the force distribution is rather heterogeneous [7]. In dense polygonal packings with finite porosity [25] and disks assemblies, [26, 27] small loads open weak contacts and hence induce a smooth transition to the anisotropy for small deviatoric loads. In all cases it is concluded that one can micromechanically characterize the anisotropic elasticity by introducing fabric coefficients, measuring the anisotropy of the contact network [11].

From the plastic part of the incremental response, dilatancy $d = -de^p/d\gamma^p$ and the stress ratio $\eta = q/p$ are related by simple linear relation $d = c(\eta - M)$ [13]. This relation is not only supported by experiments, but also it has been one of the fundamental issues in modeling the stress-strain behavior of soils. Unfortunately the theoretical assumptions of classical models (Taylor [28], Nova & Wood [20], Gutierrez & Ishihara [29]) are not verified in our model. This leads to the basic question: what lies behind of this simple stress-dilatancy relationship? Although we cannot give a definitive answer, a physical explanation would be that a granular medium close to the plasticity limit behaves like a *strange fluid*, that obeys this stress-dilatancy relation as an internal kinematical constraint. This constraint could be seen as the counterpart of the well-known incompressibility condition of fluids. In this context we should address to the existing correlation between the mean orientation of the sliding contacts and the plastic flow direction [13]. This correlation suggests that this internal constraint can be micromechanically interpreted from the induced anisotropy of the sub-network of the sliding contacts.

4 Granular ratcheting

In this last section we introduce a long time effect in granular materials, which is still under discussion in the scientific and engineering community. This effect is known as *granular ratcheting*, and it refers the constant accumulation of permanent deformation per cycle, when the granular sample is subjected to load-unload stress cycles with extremely small loading amplitudes. Although there is wide experimental evidence about accumulation of permanent deformation under cyclic loading [30–37], it is not clear whether this effect remains for small loading amplitudes, or there is a certain regime where the material behaves perfectly elastic [20, 38, 39]. It is still also not clearly understood what is the role of the micromechanical processes such as sliding, crushing and wearing of the grains, in the accumulation of plastic deformation with the number of cycles [30–33,40,41]. Here we present numerical evidence of this ratcheting effect for small loading amplitudes on assemblies of densely packed polygons. This can be detected at the micromechanical level by a ratchetlike behavior at the contacts. This effect excludes the existence of the rather questionable finite elastic regime of noncohesive granular materials.

We use samples with volume fractions lower than one. First, the polygons are placed randomly inside a rectangular frame consisting of four walls. Then, a gravitational field is applied and the sample is allowed to consolidate. The external load is imposed by applying a force $\sigma_1 H$ and $\sigma_2 W$ on the horizontal and vertical walls, respectively. Here σ_1 and σ_2 are the vertical and horizontal stresses. H and W are the height and the width of the sample. Next, the sample is isotropically compressed until the pressure p_0 is reached.

When the velocity of the polygons vanishes gravity is switched off. Then, the vertical stress $\sigma_1 = p_0$ is kept constant and horizontal stress is modulated as $\sigma_2 = p_0 + \Delta\sigma[1 - \cos(\pi t/t_0)]/2$. Part (a) of Fig. 3 shows the relation between the stress $q = (\sigma_1 - \sigma_2)/2$ and the shear strain γ in the case of a loading amplitude $\Delta\sigma = 0.424p_0$. This relation consists of open hysteresis loops which narrow as consecutive load-unload cycles are applied. This hysteresis produces an accumulation of strain with the number of cycles which is represented by γ_N in the part (b) of Fig 3. We observe that γ_N consists of short time regimes, with rapid accumulation of plastic strain, and long time *ratcheting* regimes, with a constant accumulation rate of plastic strain of around 2.4×10^{-6} per cycle. The relation between the stress and the volume fraction is shown in part (c) of Fig. 3. This consists of asymmetric compaction-dilation cycles leading to compact during the cyclic loading. This compaction is shown in part (d) of the Fig. 3. We observe a slow variation of the volume fraction during the *ratcheting* regime, and a rapid compaction during the the transition between two ratcheting regimes, whereas the slope of γ_N shows no dependency on the compaction level of the sample. The evolution of the volume ratio seems to be rather sensitive to the initial random structure of the polygons. Even so we found that after 8×10^3 cycles the volume fraction still slowly increases in all the samples, without reaching the saturation level.

By following the evolution of the contact forces one can explain this particular behavior. These forces have a convoluted spatial structure that is called force network. Even under isotropic compression, the strong heterogeneities of the force network produce a considerable amount of contacts reaching the sliding condition $|f_t| = \mu f_n$. Those sliding contacts carry most of the irreversible deformation of the granular assembly during the cyclic loading. Opening and closure of contacts are quite rare events, and the coordination number of the packing keeps it approximately its initial value 4.2 ± 0.08 in all the simulations. After certain loading cycles the contact forces reach the quasi-periodic behavior. In this regime, a fraction of the contacts reaches almost periodically the sliding condition. The load-unload asymmetry of the contact force loops makes the contacts slip the same amount and in the same direction during each loading cycle. When the ratcheting regime is reached, each particle within the packing has a certain displacement and accumulates the same rotation for each cycle. It is of great interest to study the patterns created by the rotational and displacement field of all the grains. Slip zones, rotational bearings and vortical structures persist during the long time of a ratcheting regime [18, 42]. Recently we have shown that these rotational patterns promote a strong reduction of strength and frictional dissipation in shear cells [11]. It is an open question how to introduce such rotational modes in the continuum, but it will certainly improve the understanding of the complex behavior of the granular materials.

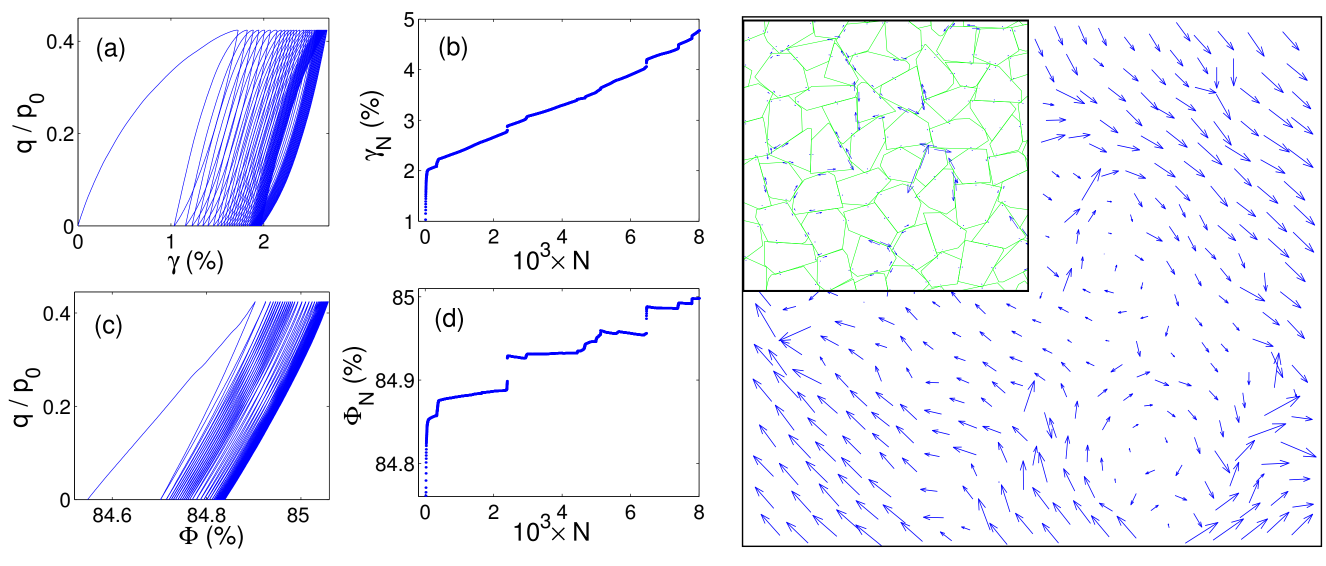

Figure 3. Left: (a) Deviatoric stress versus shear strain in the first 40 cycles. (b) permanent (plastic) strain γ_N after N cycles versus the number of cycles. (c) stress against the volume fraction in the first 40 cycles. (d) volume fraction Φ_N after N cycles versus number of cycles. Right: displacement field after one cycle in the ratcheting regime. Inset: Permanent deformation per cycle at the contacts.

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